A defensible algorithmic contribution beyond an HEO workbench

Recommended concept: event-certified, covariance-aware adaptive multifidelity propagation

The strongest opportunity is not to invent another standalone regularized integrator. Several important pieces already exist: non-averaged regularized methods have been compared with semi-analytical propagation for GTO and high-altitude HEO, with reported speed advantages for HEO at comparable accuracy.[1] Adaptive multifidelity uncertainty propagation already combines low-fidelity particles with selected high-fidelity re-propagations, and multifidelity orbit-determination work reports a twelve-fold runtime reduction without statistically significant filter-performance loss in its tested case.[2, 3] Verified interval propagation also already provides mathematically guaranteed position and velocity enclosures.[4]

A more defensible contribution is a **HEO-specific propagation controller that switches among these methods using event-time risk and covariance growth, then certifies the switch with an enclosure or independent cross-check**.

Call it **ECR-MF: Event-Certified Risk-aware Multifidelity propagation**.

What would be new

The novelty claim should be narrow and testable:

> ECR-MF is an adaptive propagator for highly eccentric Earth orbits that selects between regularized non-averaged, semi-analytical, and high-fidelity numerical propagation using a predicted state/event/covariance error budget, and triggers local high-fidelity certification when perigee, maneuver, eclipse, or covariance-risk conditions are detected.

This is more specific than "a hybrid propagator." Hybrid and multifidelity propagation already exist. The distinctive contribution is the **decision rule and certification target**: switching is driven not only by state error or runtime, but by the probability that a method switch will violate an event-time, maneuver-state, or covariance-coverage requirement.

Algorithmic design

1. Maintain three propagation modes

- **Fast mode**: semi-analytical or low-order numerical propagation for quiet portions of the orbit and large ensembles.
- **Regularized mode**: adaptive Sundman/KS or nonsingular-element propagation for high-eccentricity and perigee-dominated segments.
- **Certification mode**: high-fidelity numerical propagation with the full qualified force model, tight tolerances, and explicit event localization.

The multisegment Chebyshev–Picard literature shows why true-anomaly-informed segmentation is useful for highly elliptical motion: nonuniform nodes and multiple segments can control approximation behavior across the orbit.[5]

2. Propagate a risk state, not only a nominal state

At every segment boundary, estimate:

- predicted position and velocity error;
- perigee/apogee and maneuver event-time error;
- covariance growth and non-Gaussianity indicator;
- force-model truncation error;
- disagreement between two propagation modes;
- distance to a mission threshold.

The controller should switch to certification mode when any predicted quantity approaches its threshold. This makes the method mission-aware rather than globally "high fidelity."

3. Use event-time as a first-class error variable

For each upcoming event, propagate both the state and the event-time sensitivity. A mode switch is accepted only if the predicted event-time difference between the active method and certification mode is below the event budget. This directly addresses a known weakness of regularized propagation: event-based stopping can introduce timing or phasing errors even when the geometric trajectory is very accurate.[6]

4. Add local dual propagation at decision points

At perigee, burn start/stop, eclipse transition, and covariance inflation points, run two independent short propagations:

1. the selected low-cost method;
2. the certification method.

Use their disagreement to update an online local error model. If disagreement exceeds the local budget, roll back to the last certified state and continue in certification mode until the risk subsides.

5. Couple covariance to method selection

Do not switch methods solely on nominal-state disagreement. Propagate a small sigma-point or polynomial-chaos ensemble through the candidate modes. Escalate fidelity when the low-cost mode changes the covariance eigenstructure, tail probability, or event-time distribution beyond the mission budget. This extends existing multifidelity uncertainty propagation from "reduce ensemble runtime" to "protect operational event and covariance requirements." Existing multifidelity work provides the foundation, but not this HEO-specific event-risk rule in the retrieved evidence.[2, 3]

6. Provide a formal certificate

For every output segment, emit:

- active propagation mode;
- force-model configuration;
- local error estimate;
- event-time bound;
- covariance consistency status;
- reason for every mode switch;
- whether an independent certification run was performed.

For safety-critical segments, optionally attach interval enclosures. Verified interval propagation has already demonstrated the value of mathematically guaranteed position and velocity solutions; ECR-MF would use that idea selectively rather than paying interval-arithmetic cost over the entire arc.[4]

Why this is more valuable than a new generic integrator

A new generic integrator would compete with mature numerical and regularized methods on a crowded axis. ECR-MF instead targets the operational failure modes that matter in HEO:

- missed or mistimed perigee and maneuver events;
- long-arc phase drift hidden by good instantaneous position norms;
- covariance becoming overconfident after model switching;
- excessive runtime from applying full force fidelity to every ensemble member and every orbit segment;
- inability to explain why a particular force model or propagation mode was used.

Minimum research contribution

A publishable first version should include:

1. a formal switching policy based on state, event-time, and covariance budgets;

2. a proof or empirical guarantee that the switch does not exceed the specified local error budget;
3. Molniya and GTO test cases;
4. comparisons against semi-analytical, regularized, and full numerical baselines;
5. Monte Carlo covariance-coverage and NEES tests;
6. runtime–accuracy–event-risk Pareto curves;
7. ablation studies showing the value of event-time sensing, covariance sensing, and local dual propagation separately.

The key result should not be "ECR-MF is faster." It should be: **for the same event-time and covariance-coverage requirement, ECR-MF reduces high-fidelity propagation cost while preserving a machine-checkable certificate at the segments that matter**.

What cannot be claimed yet

A literature search cannot establish that this exact combination is globally novel. Regularization, multifidelity uncertainty propagation, adaptive force models, multisegment methods, and verified interval propagation all have prior art.[1, 2, 4, 5] The defensible path is to claim a new **HEO-specific integration and certification policy**, then perform a formal prior-art search before using "novel" in a paper, grant, or patent.

References

1. Amato D, Bombardelli C, Baù G, et al (2018) Non-averaged regularized formulations as an alternative to semi-analytical orbit propagation methods. Celestial mechanics & dynamical astronomy. https://doi.org/10.1007/s10569-019-9897-1

2. Jones B, Weisman R (2019) Multi-fidelity orbit uncertainty propagation. Acta Astronautica. https://doi.org/10.1016/J.ACTAASTRO.2018.10.023

3. Jones B (2018) Multi-Fidelity Methods for Orbit Determination

4. Geul J, Mooij E, Noomen R (2021) Verified Regularized Interval Orbit Propagation. Journal of Guidance Control and Dynamics. https://doi.org/10.2514/1.G004947

5. Kim D, Junkins J, Turner J (2015) Multisegment Scheme Applications to Modified Chebyshev Picard Iteration Method for Highly Elliptical Orbits. Mathematical Problems in Engineering. https://doi.org/10.1155/2015/290781

6. Urrutxua H, Álvarez J, Bombardelli C (2016) Impact of the integration stop-condition on the accuracy of regularized orbit formulations